

Phase-Space Regime Map (Structural Edition)

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Abstract

The Phase-Space Regime Map formalises how reasoning systems occupy and transition between structural regimes. It defines regime surfaces, regime boundaries, regime stability, collapse-driven transitions, recurrence-driven transitions, gradient-horizon coupling, and motif-induced regime shifts. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.

1 Introduction

A regime is a region of phase-space where structural behaviour is predictable. The regime map governs:

- Regime Surfaces
- Regime Boundaries
- Regime Stability
- Collapse-Driven Transitions
- Recurrence-Driven Transitions
- Gradient-Horizon Coupling
- Motif-Induced Regime Shifts

2 Regime Surfaces

Let P denote phase-space coordinates and $R(P)$ regime stability. A regime surface exists when:

$$R(P) > R_{\min}.$$

Regime radius:

$$S = f(R(P)).$$

3 Regime Boundaries

A boundary exists when:

$$R(P) = R_{\min}.$$

Boundary sharpness:

$$B = |\nabla R(P)|.$$

4 Regime Stability

Stability condition:

$$\frac{dR}{dt} > 0.$$

Stability decreases when:

- Motif load increases
- Gradient steepening occurs
- Horizon drift intersects regime surfaces

5 Collapse-Driven Transitions

Collapse condition:

$$\Delta R < 0.$$

Collapse velocity:

$$V_c = -\frac{dR}{dt}.$$

6 Recurrence-Driven Transitions

Recurrence condition:

$$\Delta R > 0.$$

Recurrence velocity:

$$V_r = \frac{dR}{dt}.$$

7 Gradient-Horizon Coupling

Coupling strength:

$$C = g(\nabla F, H).$$

8 Motif-Induced Regime Shifts

Shift strength:

$$M = h(M_1, M_2, P).$$

9 Conclusion

The Phase-Space Regime Map defines regime surfaces, boundaries, stability, transitions, and motif-driven shifts. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.